Joshua Becker

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EDUCATION

Stevens Institute of Technology | Hoboken, NJ — Expected May 2029

Bachelor of Engineering in Biomedical Engineering

GPA: 3.9 | **Awards:** A. James Clark Scholarship, Edwin A. Stevens Scholarship

Relevant Coursework: Engineering Statics, Circuits & Systems, Biomedical Engineering Techniques, Biology & Biotechnology, General Chemistry I & Lab, Differential Equations, Multivariable Calculus, AI & Machine Learning

TECHNICAL SKILLS

Programming: Python, C++

Tools & Platforms: PyTorch, SolidWorks, Arduino, PyQt5, Next.js, Slurm (HPC), Docker, Git, Linux, n8n, Ollama, MS Office

Languages: English (native), Spanish (native)

RESEARCH EXPERIENCE

Stevens Institute of Technology | Hoboken, NJ — June 2026 – August 2026

Undergraduate Researcher, Image-Based Deep Learning for Stroke Therapy (Clark Pinnacle Stipend)

- Trained a two-stage PyTorch model (U-Net segmentation → MLP regressor) that estimates guidewire contact force from camera frames with 19.1 mN mean absolute error, 3.5× better than baseline, on a 24,322-frame dataset
- Ran GPU training through Slurm on the university's NVIDIA L40S cluster; the U-Net reached 0.99 validation Dice with 7.1 ms/frame inference (4.7× faster than real time)
- Designed, built, and calibrated an automated data-collection rig (Arduino stepper stage, HX711 load cell, LED-flash video sync), modifying the HX711 for 8× faster sampling and calibrating to $R^2 = 1.000$ (0.22% max error)
- Programmed a multithreaded PyQt5 control panel (1,200+ lines) with live force plotting and ~3 ms camera–sensor sync; cut preprocessing time 6× and caught a flash-detection bug that mislabeled frames by up to 5.7 s

PROJECTS

Autonomous Tour-Guide Robot | Stevens Institute of Technology — Spring 2026

- Built a mobile robot in a 3-person team that navigated to sequenced targets via MQTT coordinates, ArUco marker localization, and three ultrasonic sensors, reaching all 4 targets in about 1.5 min
- Wrote the navigation code with a teammate (atan2 heading control, slow-down and arrival zones, wall-following around obstacles) and did all of the wiring, including a solder-free power bus verified with a multimeter

MaterialFlow — Material Tracking Web App | Stevens Institute of Technology — Spring 2026

- Built a role-based web app (Next.js, Prisma) replacing text-message material requests for electrical contractors with a tracked 4-role workflow: material list → PM approval → warehouse → purchasing
- Designed status-tracked lists (draft → submitted → approved → complete) with per-site inventory, an audit log, and AI chat summaries

Smart Watering System | Stevens Institute of Technology — Fall 2025

- Co-designed a self-watering plant device in a 3-student team, modeling a waterproof SolidWorks enclosure for an ESP32, pump, and sensors, and programming it to publish readings over MQTT for remote monitoring

LEADERSHIP & INVOLVEMENT

Clark Scholars Program | Clark Cohort Representative — Feb 2026 – Present

- Elected by peers to represent the 11-scholar Class of 2029 Clark cohort, working with program staff to plan events and passing scholar feedback back to the program
- Served as Stevens' point of contact for the Clark Scholars Summit, coordinating with Summit organizers

LAA | Ethnic Student Council (ESC) Representative — Sept 2026 – Present

- Represent LAA on the Ethnic Student Council, coordinating council-wide events with other cultural organizations
- Support club budgeting and SGA budget addendum requests, recommending funding adjustments for events
- Led an on-campus cultural event and currently coordinating a co-sponsored event with another cultural organization

Society of Hispanic Professional Engineers (SHPE) | Member, 2026 National Convention